Bilateral Teleoperation for Safe Surface-Contact Tasks

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***Abstract*—This report reviews and adapts the bilateral teleoperation method of Pagliara *et al.* The method combines Cartesian admittance control, motion-error-based virtualized force feedback, and reference saturation for safe contact with a rigid surface. We map the original haptic device and robot to a DualSense controller and a Franka arm simulated in PyBullet. Because the DualSense cannot render Cartesian force, feedback is conveyed through vibration and adaptive-trigger resistance. The baseline reproduction uses a planar inspection surface; the proposed extension adds desired normal-force tracking and online surface-normal estimation for curved surfaces.**

I. Introduction

TELEOPERATED surface inspection requires a robot to move a probe along a surface while maintaining stable contact. The operator must control the probe's tangential motion and recognize whether the contact force is insufficient, acceptable, or potentially damaging. However, directly reflecting measured contact forces can produce abrupt feedback during contact transitions. Furthermore, robot compliance alone does not prevent the contact force from increasing when the operator continues commanding motion into the surface.

To address these challenges, this project adopts the bilateral teleoperation method proposed by Pagliara *et al.* [1]. The method combines Cartesian admittance control, motion-error-based virtualized force feedback, and force-threshold reference saturation to improve contact performance and safety. In our implementation, the original robot and force-reflecting haptic device are replaced by a simulated Franka arm and a PS5 DualSense controller. The operator commands the probe's Cartesian motion using the controller, while vibration and adaptive-trigger resistance communicate the contact condition. The initial experiment will use a planar surface to reproduce the essential control method. The proposed extension will then consider curved-surface inspection with desired normal-force tracking and online surface-normal estimation.

A. Problem Addressed by the Paper

The paper considers precision contact tasks performed through bilateral haptic teleoperation. In a conventional direct force-reflection architecture, the contact force measured at the robot wrist is transmitted to the operator through the haptic device. The authors identify two main limitations:

- **Abrupt force feedback:** Measured contact forces can change suddenly when contact is established or lost. Reflecting these changes directly can push the operator's hand back abruptly, reduce task accuracy, and excite oscillations in the teleoperation loop.
- **Unbounded contact-force growth:** An admittance or impedance controller makes the robot compliant, but it does not necessarily limit the contact force. The force may continue to increase if the operator keeps commanding motion into the surface.

The proposed architecture addresses these limitations using three elements:

1) **Cartesian admittance control:** The slave robot responds compliantly to the difference between the desired and measured interaction forces.
2) **Virtualized force feedback:** The feedback rendered to the operator is computed from the admittance-controller motion error using a virtual spring–damper model instead of directly using the force-sensor measurement. Because the motion error varies more smoothly than the measured force, the resulting feedback is less abrupt. It can also be tuned independently of the slave robot's interaction dynamics.
3) **Reference saturation:** When the measured contact force exceeds a prescribed threshold, the affected component of the motion reference is held constant. This prevents additional commanded penetration and limits further force growth.

II. Original System and Interfaces

A. Human–Master Side

The original master station uses a 3D Systems Touch device to measure a six-degree-of-freedom stylus pose and render three-degree-of-freedom force feedback. Its displacement from the initial pose is mapped to the robot end-effector using a 1:1 position scale.

Let H_b and H_e denote the haptic-device base and stylus frames, respectively, and let R_b and R_e denote the robot base and end-effector frames. The orientation and position mappings are

$$\boldsymbol{R}_{R_e}^{R_b} = \boldsymbol{R}_{H_b}^{R_b}\boldsymbol{R}_{H_e}^{H_b}\boldsymbol{R}_{R_e}^{H_e}, \tag{1}$$

$$\boldsymbol{p}_{R_e}^{R_b} = \boldsymbol{p}_{R_e}^{R_b}(0) + \boldsymbol{R}_{H_b}^{R_b}\left(\boldsymbol{p}_{H_e}^{H_b} - \boldsymbol{p}_{H_e}^{H_b}(0)\right). \tag{2}$$

Here, $\boldsymbol{R}_{R_e}^{H_e}$ specifies the fixed alignment between the stylus and robot end-effector frames. The initial positions are denoted by $\boldsymbol{p}_{R_e}^{R_b}(0)$ and $\boldsymbol{p}_{H_e}^{H_b}(0)$.

To reduce hand tremor, the mapped position is filtered using a moving average. For a maximum window of n samples,

$$\boldsymbol{p}_h^{R_b}(k) = \frac{1}{m_k} \sum_{\ell=0}^{m_k-1} \boldsymbol{p}_{R_e}^{R_b}(k-\ell), \qquad m_k = \min(k+1, n). \tag{3}$$

The filtered position $\boldsymbol{p}_h^{R_b}$ becomes the human-generated robot reference. The orientation is filtered separately using quaternion averaging.

B. Slave–Environment Side

The original slave is a position-controlled UR10 carrying a chalk tool and a Robotiq FT150 six-axis force/torque sensor. Tool-payload gravity is compensated. The operator writes A–C–G on a wall-mounted blackboard. Cartesian admittance generates the compliant command, and the environment is approximated locally as a linear spring for the force-limitation derivation.

III. Control Architecture

A. Cartesian Admittance Control

Let $\boldsymbol{p}_d^{R_b}$ denote the desired position and $\boldsymbol{p}_c^{R_b}$ the compliant position command. Their difference, expressed in the end-effector frame, is

$$\widetilde{\boldsymbol{p}}^{R_e} = \boldsymbol{R}_{R_b}^{R_e} \left(\boldsymbol{p}_d^{R_b} - \boldsymbol{p}_c^{R_b} \right). \tag{4}$$

The desired interaction dynamics are

$$\boldsymbol{M}_d \ddot{\widetilde{\boldsymbol{p}}}^{R_e} + \boldsymbol{K}_D \dot{\widetilde{\boldsymbol{p}}}^{R_e} + \boldsymbol{K}_P \widetilde{\boldsymbol{p}}^{R_e} = -\widetilde{\boldsymbol{f}}^{R_e}, \tag{5}$$

where

$$\widetilde{\boldsymbol{f}}^{R_e} = \boldsymbol{f}_d^{R_e} - \boldsymbol{f}_e^{R_e}. \tag{6}$$

The matrices $\boldsymbol{M}_d$, $\boldsymbol{K}_D$, and $\boldsymbol{K}_P$ are diagonal virtual mass, damping, and stiffness matrices. The vector $\boldsymbol{f}_e^{R_e}$ is the force exerted by the robot on the environment and is equal and opposite to the measured sensor force.

The paper uses $\boldsymbol{f}_d^{R_e} = \boldsymbol{0}$. Therefore, the controller provides compliant behavior rather than tracking a nonzero desired force. Using the error definition in (4), the compliant command is

$$\boldsymbol{p}_c^{R_b} = \boldsymbol{p}_d^{R_b} - \boldsymbol{R}_{R_e}^{R_b} \widetilde{\boldsymbol{p}}^{R_e}. \tag{7}$$

B. Interaction-Force Limitation

The environment is approximated by a linear spring with stiffness $\boldsymbol{K}_e$ and rest position $\boldsymbol{p}_e$. At steady state,

$$\boldsymbol{f}_e^{R_e} = \boldsymbol{K}_{eq} \left(\boldsymbol{p}_e^{R_e} - \boldsymbol{p}_d^{R_e} \right), \tag{8}$$

where, for diagonal stiffness matrices,

$$\boldsymbol{K}_{eq} = (\boldsymbol{K}_P \boldsymbol{K}_e)(\boldsymbol{K}_P + \boldsymbol{K}_e)^{-1}. \tag{9}$$

The jth component of the desired reference is saturated when the corresponding contact-force component exceeds its threshold:

$$p_{d,j}^{R_b} = \begin{cases} p_{h,j}^{R_b}, & |f_{e,j}^{R_b}| < f_{th,j}, \\ \bar{p}_{d,j}^{R_b}, & |f_{e,j}^{R_b}| \geq f_{th,j}. \end{cases} \tag{10}$$

The saturation value is estimated as

$$\bar{p}_{d,j}^{R_b} = p_{e,j}^{R_b} + \frac{f_{th,j}}{K_{eq,j}}, \tag{11}$$

where $p_{e,j}^{R_b}$ is the estimated surface rest position. In the original implementation, it is approximated using the last compliant position commanded when saturation is activated. This mechanism prevents unbounded force growth after the threshold is crossed. However, it does not guarantee that the transient force remains below $f_{th,j}$ because the environmental model is approximate and saturation is inactive before the threshold is detected.

C. Virtualized Force Feedback

Instead of directly rendering the measured force, the paper forms a virtual spring–damper force from the unsaturated admittance error:

$$\boldsymbol{f}_h^{H_b} = \boldsymbol{K}_h \widetilde{\boldsymbol{p}}_h^{H_b} + \boldsymbol{D}_h \dot{\widetilde{\boldsymbol{p}}}_h^{H_b}, \tag{12}$$

where

$$\widetilde{\boldsymbol{p}}_h^{H_b} = \boldsymbol{R}_{R_b}^{H_b} \left(\boldsymbol{p}_h^{R_b} - \boldsymbol{p}_c^{R_b} \right), \tag{13}$$

$$\dot{\widetilde{\boldsymbol{p}}}_h^{H_b} = \boldsymbol{R}_{R_b}^{H_b} \boldsymbol{R}_{R_e}^{R_b} \dot{\widetilde{\boldsymbol{p}}}^{R_e}. \tag{14}$$

The error in (13) uses the operator's unsaturated reference $\boldsymbol{p}_h^{R_b}$ rather than the limited reference $\boldsymbol{p}_d^{R_b}$. Consequently, the feedback continues to represent the operator's attempt to move farther into the environment even when robot motion has been limited. Because $\widetilde{\boldsymbol{p}}_h$ is generated by a second-order system, it changes more smoothly than the measured contact force. The matrices $\boldsymbol{K}_h$ and $\boldsymbol{D}_h$ can therefore be adjusted independently of the robot-side admittance parameters.

D. DualSense Feedback Adaptation

The DualSense cannot apply a Cartesian force to the operator. Therefore, the computed virtual force is converted into vibration amplitude and adaptive-trigger resistance.

Let $\boldsymbol{n}$ denote the unit surface normal and define

$$f_{h,n} = \boldsymbol{n}^T \boldsymbol{f}_h, \qquad f_{e,n} = \boldsymbol{n}^T \boldsymbol{f}_e. \tag{15}$$

The proposed feedback mapping is

$$\begin{aligned} a_{\mathrm{vib}} &= \mathrm{sat}_{[0,1]} \left(\frac{|f_{h,n}|}{f_{\mathrm{vib,max}}} \right), \\ a_{\mathrm{trig}} &= \mathrm{sat}_{[0,1]} \left(\frac{\max(0, |f_{e,n}| - f_{\mathrm{warn}})}{f_{\mathrm{stop}} - f_{\mathrm{warn}}} \right). \end{aligned} \tag{16}$$

Vibration communicates the virtualized-feedback magnitude, while trigger resistance warns the operator that the measured normal force is approaching the stopping threshold. This adaptation preserves the feedback variable used by the

TABLE I
RESULTS REPORTED BY THE PRIMARY PAPER [1].

Scenario	Success	MD (N)	$\Delta f_{\max}$ (N)
A	3rd attempt	48.32	80.40
B	2nd attempt	54.88	137.21
C	1st attempt	1.40	14.22

paper but changes its physical presentation from kinesthetic force reflection to tactile sensory substitution. It must therefore be reported as an adaptation rather than an exact reproduction.

In the measurement-based baseline, the vibration channel uses $|f_{e,n}|$ instead of $|f_{h,n}|$. This preserves the distinction between the paper's direct and virtualized feedback conditions without claiming that the DualSense reproduces a Cartesian force vector.

E. Paper Baselines and Reported Metrics

The paper compares three architectures. Scenario A uses measurement-based force feedback without reference saturation. Scenario B uses virtualized feedback without saturation. Scenario C uses virtualized feedback with saturation. Writing quality is assessed qualitatively through stroke accuracy and continuity. Quantitative metrics compare the normal-force profiles of robot and manual writing:

$$MD = |\bar{f}_{\mathrm{hum}} - \bar{f}_{\mathrm{rob}}|, \qquad \Delta f_{\max} = |f_{\max,\mathrm{hum}} - f_{\max,\mathrm{rob}}|. \tag{17}$$

These results support the paper's safety and task-performance claims, but they come from a small number of executions rather than a controlled, multi-participant user study.

IV. Required Signals and Assumptions

Table II summarizes the signals required by the original architecture and how they will be obtained using the Franka–PyBullet platform.

The reproduction assumes that the surface is locally rigid and that its stiffness is bounded and approximately known. This information is required to calculate the reference-saturation value. The measured contact force must be filtered, and all positions, velocities, and forces must use consistent coordinate frames and sign conventions. Numerical integration of the admittance dynamics must remain stable at the selected simulation time step.

Because the original paper does not evaluate communication delay, delay compensation is not included in the reproduced controller. Nevertheless, the simulation period, controller execution time, and DualSense input latency will be measured and logged so that their influence is not ignored.

V. Block Diagram

Figure 1 shows the architecture planned for the project. It preserves the main control structure of the primary paper while replacing the Touch haptic device and UR10 robot with a DualSense controller and a Franka arm simulated in PyBullet. The numbered markers correspond to the explanations in Table III.

VI. Literature Review

To place the selected method in context, we reviewed research from four areas that influence our project. These include stability in bilateral teleoperation, haptic interfaces and sensory substitution, teleoperated surface-treatment systems, and force-control methods for surfaces with uncertain geometry. Table IV compares the primary paper with the ten supporting studies and highlights how each study contributes to our proposed system.

A. Bilateral Teleoperation and Stability

Hokayem and Spong [2] identify stability, transparency, communication delay, and master–slave scaling as central challenges in bilateral teleoperation. Nuño *et al.* [3] explain how passivity can maintain stable energy exchange among the human operator, master device, slave robot, and remote environment. These studies provide the theoretical context for the selected paper, which improves feedback behavior but does not provide a formal stability analysis.

Santiago *et al.* [4] propose a variable $P+d+f$ controller for delayed bilateral teleoperation. Their method adjusts damping using communication delay and a feedback-power signal. Although delay compensation is outside the initial reproduction, this work provides a suitable reference for a future network-delay extension.

B. Haptic Interfaces and Sensory Substitution

Bong *et al.* [5] develop a three-degree-of-freedom motorized haptic interface using virtual spring–damper feedback. Their system demonstrates a complete bilateral position-and-force loop, but its main contribution depends on custom force-reflecting hardware that is unavailable in this project.

Motamedi *et al.* [6] compare visual, vibrotactile, and pressure feedback during robotic grasp, contact, and slip-detection tasks. Although their manipulation task differs from surface inspection, the study supports tactile sensory substitution and comparative user-study conditions when true Cartesian force reflection is unavailable.

C. Teleoperated Surface Treatment

Girbés-Juan *et al.* [7] study haptic and visual assistance for dual-arm teleoperation in surface-conditioning tasks. Their evaluation includes contact behavior, completion time, and surface coverage, making it the closest application-level reference for our project. However, direct reproduction would require a dual-arm platform.

Their expanded study [8] combines haptic feedback, visual assistance, and inertial motion capture for remote control of a dual-arm robot. The reported surface-conditioning experiments support our selection of task time, path coverage, accuracy, and operator workload as evaluation measures.

González *et al.* [9] combine augmented virtuality, haptic feedback, perception, and tool-orientation assistance for industrial surface treatment. The study demonstrates the value of combining human adaptability with robot-side assistance, but its perception and visualization system is broader than the scope of this project.

TABLE II
REQUIRED CONTROL SIGNALS AND THEIR REALIZATION IN THE PROJECT.

Signal	Original source	Franka–PyBullet realization
Human motion command	Haptic-stylus pose	DualSense stick and button inputs mapped to Cartesian velocity commands, with a button used as a clutch.
Human reference p_h	1:1 pose mapping followed by moving-average filtering	Numerical integration of the filtered Cartesian velocity command from the initial end-effector pose.
End-effector state	Slave-robot pose and velocity	PyBullet end-effector link pose and velocity expressed in the robot base frame.
Contact wrench f_e	Six-axis wrist force/torque sensor	Probe–surface contact forces reconstructed from PyBullet contact data and transformed into the selected control frame.
Surface contact state	Force threshold and estimated environmental rest position	Contact-point detection, estimated surface normal, and filtered normal contact force.
Compliant command p_c	Cartesian admittance controller	Numerical integration of the admittance dynamics followed by inverse kinematics.
Joint command q_{cmd}	Robot low-level position controller	Damped least-squares inverse kinematics followed by PyBullet joint-position control.
Feedback variable f_h	Virtual spring–damper model	Virtual-feedback calculation using the unsaturated error $p_h - p_c$.
Rendered feedback	Cartesian force applied by the haptic device	DualSense vibration and adaptive-trigger resistance.
Visual feedback	Remote camera view	PyBullet camera view with the probe, target path, and optional contact-force indicator.

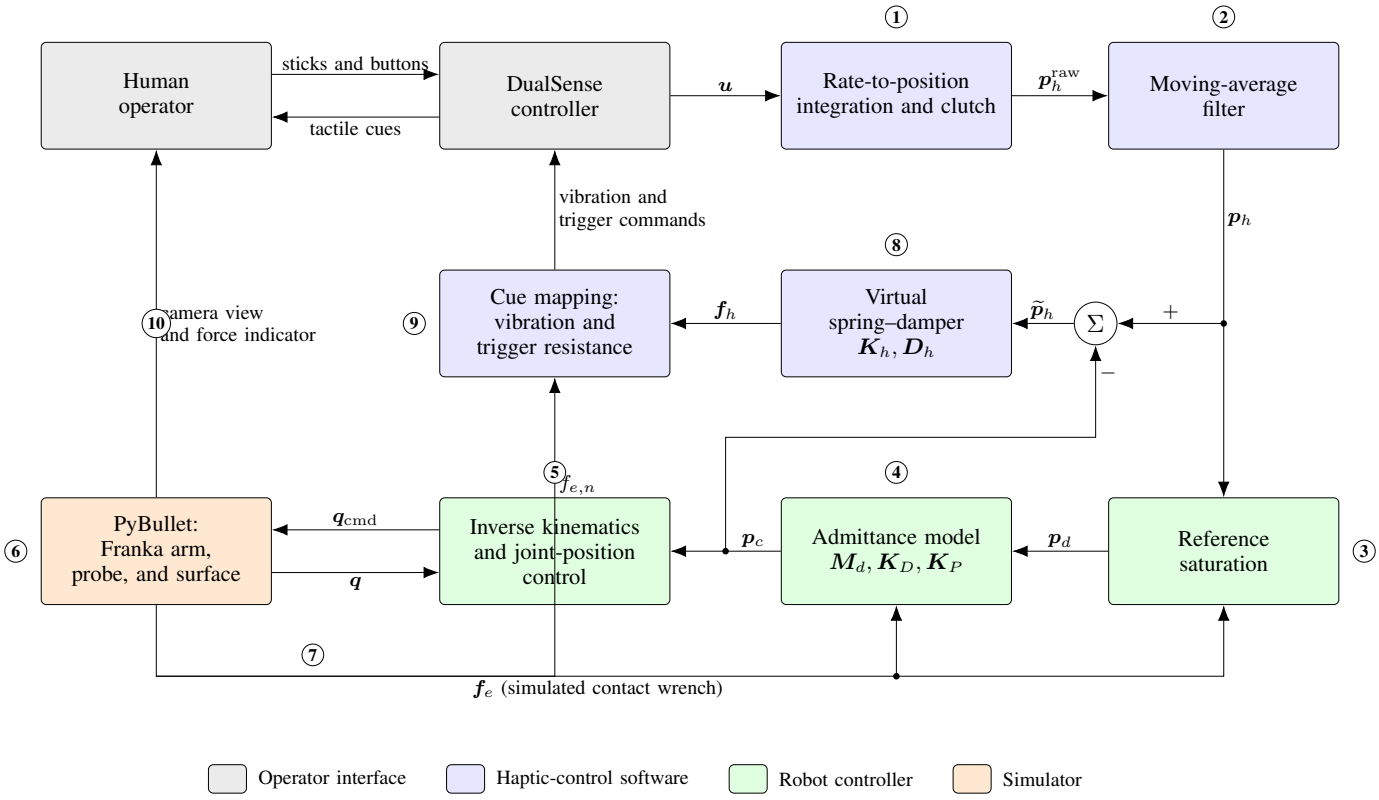


Fig. 1. Planned bilateral teleoperation architecture. The forward path converts DualSense inputs into a compliant Franka command. Return paths send the contact wrench to the admittance, saturation, and trigger channels; virtualized feedback to the vibration channel; and the camera view to the operator.

D. Force Control for the Surface Extension

Nasiri and Wang [10] propose a hybrid force/motion controller that estimates the surface normal online from measured force and commanded velocity while compensating for tangential friction. The method does not include bilateral teleoperation, but it provides the main technical basis for extending our experiment from a planar surface to an uncertain curved surface.

Solanes *et al.* [11] present a cooperative surface-treatment system in which the robot maintains the desired contact pressure and tool orientation while a human guides the tangential motion. Their adaptive sliding-mode controller is more complex than the selected admittance controller, but the work supports separating normal-force regulation from tangential human commands. It also motivates force-tracking and path-following errors as evaluation metrics.

Overall, the supporting literature confirms that the selected paper offers a suitable balance between relevance and reproducibility. Its admittance controller, virtualized feedback, and reference-saturation mechanism can be implemented without

TABLE III
ANNOTATIONS FOR THE CONTROL ARCHITECTURE IN FIG. 1.

No.	Annotation
1	DualSense stick input $\boldsymbol{u}$ is scaled and integrated to form a virtual master position. This replaces the paper's 1:1 stylus mapping in (1) and (2). A controller button provides clutch and reset functions.
2	The moving-average filter in (3) smooths the generated reference. Its output $\boldsymbol{p}_h$ represents the operator's unsaturated motion command.
3	Reference saturation follows (10). The controller passes $\boldsymbol{p}_h$ while the normal contact force remains below f_{th}. Once the threshold is reached, the normal reference is held at the value defined by (11). This block is disabled in Scenarios A and B.
4	The Cartesian admittance model in (5) uses $\boldsymbol{f}_d = \boldsymbol{0}$ to generate the compliant command $\boldsymbol{p}_c$.
5	Inverse kinematics converts the compliant Cartesian command into Franka joint-position commands $\boldsymbol{q}_{\mathrm{cmd}}$. Joint-state feedback $\boldsymbol{q}$ is obtained from PyBullet.
6	PyBullet simulates the Franka arm, inspection probe, rigid surface, robot state, probe–surface contact, and visual scene.
7	The simulated contact wrench $\boldsymbol{f}_e$ is reconstructed from the probe–surface contacts, filtered, and transformed into the selected control frame.
8	The virtual spring–damper in (12) generates $\boldsymbol{f}_h$ from the unsaturated error $\boldsymbol{p}_h - \boldsymbol{p}_c$. In Scenario A, the vibration channel instead uses the measured normal force.
9	The DualSense provides tactile sensory substitution according to (16). The virtualized force determines vibration intensity, while the measured normal contact force determines adaptive-trigger resistance.
10	A fixed PyBullet camera provides the same visual feedback in every experimental condition. A force indicator may be displayed when visual force information is part of the evaluated condition.

custom haptic hardware. The DualSense interface is treated as sensory substitution, while online surface-normal estimation provides a distinct extension beyond the original planar experiment.

VII. REPRODUCTION AND EXTENSION PLAN

A. Repeatable Task Environment

The Phase 3 environment will initially use a fixed rigid plane, a rounded probe, a visible tracing path, and a known normal direction. The operator will move the probe tangentially while managing contact. Reset will return the Franka to a collision-free joint configuration and place the probe at a fixed stand-off from the path start. Success requires completing the path while remaining within prescribed force and workspace limits. Failure includes excessive force, prolonged loss of contact, joint-limit violation, or timeout.

B. Experimental Conditions

1) **Baseline:** Direct Cartesian teleoperation with visual information only.
2) **Scenario A replication:** Admittance control with direct force-to-vibration feedback and no reference saturation.
3) **Scenario B replication:** Admittance control with virtualized motion-error feedback and no reference saturation.
4) **Scenario C/paper method:** Admittance control with virtualized feedback and reference saturation.
5) **Extension:** Scenario C on a curved surface with desired normal-force tracking and online normal estimation.

The minimum Phase 4 claim will compare Scenarios A–C. The visual-only baseline provides a simple Phase 3 reference. The curved-surface controller is explicitly separated as an extension.

C. Metrics and Hypotheses

Objective metrics will include completion time, success rate, RMS normal-force error, peak force, time above the safety threshold, contact-loss count and duration, path error, surface coverage, command smoothness, and saturation activity. PyBullet's known geometric normal will permit direct measurement of normal-estimation error in the extension.

The main hypotheses are:

- H_1: Virtualized feedback reduces high-frequency feedback variation and operator-command oscillation relative to direct force feedback.
- H_2: Reference saturation reduces peak force, excessive-force duration, and task failures relative to the same feedback without saturation.
- H_3: Haptic sensory substitution reduces force error and contact loss relative to visual-only operation without substantially increasing completion time.
- H_4 (extension): Online normal estimation reduces normal-force and path error on curved surfaces relative to a fixed-normal controller.

D. User-Study Outline

A within-subject design will expose each participant to the same task under counterbalanced feedback and control conditions after standardized practice. Objective logs will be combined with a short workload and usability questionnaire. The study will avoid claiming true Cartesian force reflection: participants receive tactile cues representing the paper's virtual-feedback signal.

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TABLE IV
COMPARISON OF THE PRIMARY PAPER AND TEN SUPPORTING STUDIES.

Work	Publisher/venue	Main contribution	Use in this project	Main limitation
Pagliara *et al.* [1]	IEEE CASE 2024	Admittance control, virtualized feedback, and reference saturation	Primary reproduction architecture	No formal stability proof or controlled user study; some implementation details are missing
Hokayem and Spong [2]	Elsevier *Automatica*, 2006	Foundational bilateral-teleoperation survey	Stability–transparency framework	Survey rather than an implementation
Nuño *et al.* [3]	Elsevier *Automatica*, 2011	Passivity-based bilateral control	Energy and stability interpretation	Not specific to surface contact
Santiago *et al.* [4]	Springer IJDC, 2024	Delay-aware variable $P+d+f$ control	Reference for a future delay extension	Not specific to surface tasks
Bong *et al.* [5]	Springer *Microsystem Technologies*, 2022	Powered three-DoF haptic master	Bilateral feedback and virtual-coupling reference	Requires custom master hardware
Motamedi *et al.* [6]	IEEE ICRA 2016	Comparison of visual, vibrotactile, and pressure feedback	Sensory-substitution and user-study design	Grasp and slip task rather than surface inspection
Girbés-Juan *et al.* [7]	IEEE *Transactions on Haptics*, 2021	Haptic and visual assistance for surface conditioning	Closest task and metrics reference	Requires a dual-arm platform
Girbés-Juan *et al.* [8]	Springer JMMUI, 2022	Multimodal dual-arm surface teleoperation	Evaluation procedure and workload measures	Requires motion capture and dual-arm hardware
González *et al.* [9]	Elsevier JMS, 2021	Augmented-virtuality surface teleoperation	Industrial task and shared-assistance reference	Large perception and visualization scope
Nasiri and Wang [10]	IEEE UR 2024	Online surface-normal estimation and friction compensation	Curved-surface extension	No human interface or haptic return path
Solanes *et al.* [11]	Elsevier *ISA Transactions*, 2018	Human-guided surface treatment with automatic force control	Normal-force regulation and surface metrics	Two force sensors and complex sliding-mode control

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